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MPJ LAB 04

Aim :- WAP to showcase the use of File I/O Exception Handling in Java

Objectives :-
1. To study I/O : stream classes.
2. Understand the different types of exceptions
3. Exception Handling using various Exception classes

Theory :-
1) Java I/O : input-output stream classes
→ Java I/O is a collection of classes and streams in java.io package that handles reading data from sources and writing data to ~~source~~ destination
1. Input Stream: Used to read data that must be taken as an input from a source array or file or any peripheral device
For eg: FileInputStream, BufferedInputStream, ByteArrayInputStream etc.
2. Output Stream: Used to write data as outputs into an array or file or any output peripheral device.
For eg: FileOutputStream, BufferedOutputStream etc.
2) Exception Handling in Java
→ It is a mechanism used to handle runtime errors so that the normal flow of the program can continue without crashing.
→ Handles abnormal conditions that occur during program execution
→ We use try-catch block for handling exceptions
3. Types of Exception classes
1. Checked exceptions: Exceptions checked by the compiler at compile time. Handle them using try-catch or throws
2. Unchecked exceptions: Not checked by compiler at compile time.

They include runtime exceptions and errors.

Platform: Open Source Java Programming tool like Eclipse Editor

Conclusion: Thus studied the exception handling concepts in java and implemented usage of I/O stream classes.

FAQs:

Q1) What is the difference between error and exception?

Ans

Exception	Error
→ Are of type java.lang.Exception	→ Are of type java.lang.Error
→ Include both checked and unchecked type	→ All errors in java are unchecked type
→ Are recoverable	→ Are not recoverable
→ Can happen at runtime and compiletime both	→ Happens at runtime

Q2.) Write a few examples for checked and unchecked exceptions?

Ans Checked Exceptions

Ex - IOException, FileNotFoundException, SQLException etc.

Eg-
```
try {
    FileReader file = new FileReader("test.txt");
} catch (FileNotFoundException e) {
    System.out.println("File not found");
}
}
```

Unchecked Exceptions

Ex - Arithemetic Exception, NullPointerException, ArrayIndexOutofBoundsExcepti

Eg
```
class main {
```

```
    public static void main(String[] args) {
        int a = 10/0;   //Arithmetic Exception
    }
}
```

Q3 What are the different keywords in Exception Handling in Java?

Ans a) try - contains code that might throw an exception
b) catch - handles exception if it occurs
c) throw - Used to explicitly throw a single exception
d) throws - Declares exceptions that a method might throw, informing caller to handle them
e) finally - Contains code guaranteed to execute after try & catch blocks regardless of whether exception was thrown, caught or handled.

Q4) What is the difference between throws and throw keywords in java?

Ans

throw	throws
→ Used to explicitely throw an exception	→ Used to declare an exception
→ followed by an instance	→ followed by class
→ Cannot throw multiple exceptions	→ You can declare multiple exceptio
→ Used inside method body ~~only~~	→ Used in method declaration

Q5) List down various byte and character stream classes in java.

Ans → Byte Stream
a) Input Stream - FileInputStream, BufferedInputStream, DataInputStream, ByteArrayInputStream etc.
b) Output Stream - FileOutputStream, BufferedOutputStream, DataOutputStream, ByteArrayInputStream etc
→ Character Stream

a) InputCharacterStream - FileReader, BufferedReader, InputStreamReader etc

b) OutputCharacter Stream - FileWriter, BufferedWriter, OutputStreamWriter etc